

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, computed from the open-interest grid, with the drift coupling motivated by Gârleanu–Pedersen–Potoshman [21] demand pressure. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable χ_t —and reduces to standard time-dependent Heston [27] at zero coupling. We work out the Hopf bifurcation of the deterministic skeleton, with first Lyapunov coefficient ℓ_1 and a Khasminskii-style stochastic shift Λ , and we pre-register the evaluation protocol before touching any data: a direct, model-free dealer-gamma (GEX) regression as the primary test, with a sliced-Wasserstein-2 surface metric and a critical-slowness-down detector. For a representative dimensionless calibration we obtain $\kappa^* \approx 0.8964$, $\omega^* \approx 0.5724$ rad/yr, and $\ell_1 \approx -0.025 < 0$ (supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$). The 2D reduction has an upper-triangular Jacobian and admits only saddle-node onset, recovering the discrete-time recursion of Dai [15]; the 3D extension is what unlocks the Hopf, distinguishing this work from Halperin and Itkin [23]’s Marketron quasi-particle SDE. For log-normal open-interest in moneyness we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space. We additionally close three theoretical loose ends: (i) a codim-2 Bautin/Bogdanov–Takens analysis showing the BT locus is empty in the canonical scan window (Theorem 2); (ii) the McKean–Vlasov mean-field correction to the single-dealer threshold, regime-dependent in sign ($\kappa_{\text{MV}}^*/\kappa_{\text{single}}^* < 1$ strictly at the canonical short-gamma regime, with frozen-dealer limit $8/21$ and first-order asymptote $1 - 0.664/\theta_G + O(\theta_G^{-2})$; Theorem 3); and (iii) a Hawkes–SV criticality correspondence placing Hardiman et al. [24]’s empirical $n \approx 1$ at the real-eigenvalue (saddle-node) stratum and the model’s Hopf as the strictly stronger oscillatory cell beyond any scalar branching ratio (Theorem 4).

Keywords

Reflexivity, dealer gamma, Hopf bifurcation, stochastic volatility, pre-registration, sliced Wasserstein, RL for finance

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [24], who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros [45] and Bouchaud [10] are the conceptual antecedents; Filimonov and Sornette [19] formalised the reflexivity-quantification programme that Hardiman et al. [24]

executed. We ask what continuous-time mechanism produces this critical reflexivity, and how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, χ_t) with explicit dealer-gamma feedback computed from the option open-interest grid, the coupling motivated by Gârleanu et al. [21] demand pressure. The variance backbone is Heston [27]; the memory equation $\dot{\chi} = -\alpha\chi + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion, with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 9–10).
- A numerical phase diagram in (ξ, ρ, σ_a) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (a direct dealer-gamma regression as primary test, sliced- W_2 over 21-day surface windows, κ -sensitivity slope, and a critical-slowness-down detector) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. [12] adapted for computational finance.

Closest precedents. Halperin and Itkin [23] (Marketron) is the nearest continuous-time, options-aware model with a memory channel, but its memory is a latent stochastic factor coupled through a nonlinear potential, ours is a deterministic price filter, and it carries no bifurcation analysis. Dai [15] share the dealer-gamma axis but produce a saddle-node onset. He et al. [25] (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents [14, 26] use delay-differential heterogeneous-belief models without an options-grid feedback channel; Brock et al. [11]’s “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker bifurcations. The hedger-flow-feedback foundations are Frey and Stremme [20], Platen and Schweizer [40], Sircar and Papanicolaou [44].

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and χ_t a memory variable encoding low-pass-filtered price history.

The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, \chi_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma \chi_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$d\chi_t = (-\alpha \chi_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

with $d\langle W^S, W^v \rangle = \rho dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, β dimensionless, and G the aggregate dealer-gamma exposure derived from the OI grid:

$$G(S, \chi, v) = \sum_{K,T} q_{K,T}(\chi) \Gamma_{K,T}(S, v) \text{sign}(K, T), \quad (4)$$

following the SqueezeMetrics dealer-net sign convention [46]. G is treated as a general C^3 functional of (S, χ, v) : trend-dependent open interest $q_{K,T}(\chi)$ gives $G_\chi \neq 0$, while a static grid gives $G_\chi = 0$ (the closed-form case below). When $\kappa = \gamma = 0$ the system collapses to standard Heston [27] (verified numerically). For variance we use the full-truncation Euler scheme of Lord et al. [35].

Why three states. A 2D specification (drop χ_t and the $\gamma\chi$ term) has a Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues: only saddle-node / transcritical bifurcations are admissible. To Hopf, the feedback must flow in both directions; the leverage channel $\gamma\chi$ together with the auxiliary memory equation (3) is the minimal structural modification that supplies it. Here χ is a deterministic moving average of log-price, distinct from the latent stochastic memory of Marketron [23], whose memory carries its own Brownian motion through a nonlinear potential.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria, Jacobian, Liu criterion

Drop the Brownian terms and work in log-deviation variables $(y, u, \chi) = (\log(S/S^*), v - \theta_v, \chi - \chi^*)$ around a locally unique equilibrium (S^*, v^*, χ^*) defined by the three stationarity conditions

$$\mu - \frac{1}{2}\sigma^2(S^*, v^*) + \kappa G(S^*, \chi^*, v^*) = 0, \quad \kappa_v(\theta_v - v^*) + \gamma\chi^* = 0, \quad \chi^* = \frac{\beta}{\alpha} \log\left(\frac{S^*}{S_0} + \frac{1}{2}v\right) T_{\text{eff}}. \quad (5)$$

We normalise $S_0 = S^*$ (a choice of origin for χ , no loss of generality), giving $\chi^* = 0$ and $v^* = \theta_v$ exactly. The deterministic skeleton becomes $\dot{x} = J(\kappa)x$ with

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_\chi \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (6)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$. The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with $c_2 = -a + \kappa_v + \alpha$, $c_1 = -a\kappa_v - \alpha\alpha + \kappa_v\alpha - \kappa G_\chi\beta$, $c_0 = -a\kappa_v\alpha - \beta(\kappa G_\chi\kappa_v + b\gamma)$. A 3D system Hopfs iff (Liu's criterion):

$$c_2 > 0, \quad c_0 > 0, \quad H(\kappa) := c_1c_2 - c_0 = 0, \quad H'(\kappa) \neq 0. \quad (7)$$

The Hopf threshold is $\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2, c_0 > 0\}$. At κ^* , J has imaginary pair $\lambda_\pm = \pm i\omega^* = \pm i\sqrt{c_1(\kappa^*)}$ plus real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.2 Theorem 1

THEOREM 1 (HOPF BIFURCATION IN THE GAMMA-COUPLED SV SKELETON). *Let the deterministic system (1–3) satisfy: (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below by a positive constant; (A2) a locally unique equilibrium for every $\kappa \in [0, \kappa_{\max}]$ with non-degenerate linearisation (multiple equilibria may exist globally; analysis is local); (A3) $\kappa \mapsto (a(\kappa), G_\chi, G_v)$ in C^3 ; (A4) there exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (7) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$; (A5) $\ell_1(\kappa^*) \neq 0$ [32, eq. 3.20]. Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^*$: (1) for κ in a one-sided neighbourhood of κ^* there is a unique smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude $\propto \sqrt{|\kappa - \kappa^*|}$; (2) orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical); (3) on Γ_κ , $u_t = v_t - \theta_v$ oscillates with amplitude $\propto \sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

PROOF SKETCH. The eigenvalue pair $\lambda_\pm(\kappa)$ is smooth near κ^* by the implicit function theorem on P . Routh–Hurwitz gives $\text{Re } \lambda_\pm(\kappa^*) = 0$, and $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_\pm \neq 0$. The centre manifold theorem [32, Thm. 5.4] gives a 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$; conjugation to Poincaré normal form and Kuznetsov [32, Thm. 3.3] yield (1)–(3). $\square$

Numerical anchor. For $\{G_x, G_v, G_\chi\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$ (constant-vol surrogate): $\kappa^* = 0.8964$, $\omega^* = 0.5724 \text{ rad/yr}$, $\lambda_3 = -2.052$, $\ell_1 = -2.53 \times 10^{-2}$ (supercritical). Integrating the deterministic skeleton at $\kappa = 1.05\kappa^*$ from random initial conditions, the trajectory converges to a closed orbit whose measured period $T_{\text{measured}} = 10.561 \text{ yr}$ matches the linear prediction $2\pi/\omega^* = 10.977 \text{ yr}$ to 3.79%—within the leading-order $O(\kappa - \kappa^*)$ Hopf-normal-form correction.

3.3 Closed-form κ^* and ℓ_1 for log-normal OI

For log-normal open-interest in log-strike, $q(\log K) \propto \exp(-(\log K - \mu_q)^2 / (2\sigma_q^2))$, the Gaussian-product identity collapses the OI integral to

$$G(a, v) = C(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (8)$$

with $C(v) = \kappa_v e^{-q_{\text{div}} T_{\text{eff}}} / \sqrt{2\pi\tau^2(v)}$, $\tau^2(v) = \sigma_q^2 + v T_{\text{eff}}$, $m(v) = \mu_q - \frac{1}{2} \log\left(\frac{S^*}{S_0} + \frac{1}{2}v\right) T_{\text{eff}}$. With $G_\chi = 0$ and $\sigma^2 = v$, $H(\kappa) = c_1c_2 - c_0$ degree-collapses from cubic to quadratic; writing $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, the Hopf threshold is

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{D}}{2 G_y^2 A}, \quad D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - \frac{L}{2}). \quad (9)$$

The discriminant $D < 0$ implies no Hopf at any κ and unconditional stability of the deterministic skeleton.

The third-order partials of (8) are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2 \langle p, B(q, J^{-1} B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1} B(q, q)) \rangle \right], \quad (10)$$

with $Jq = i\omega^* q$, $J^\top p = -i\omega^* p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term closed-form rational in the G -partials and $(\kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50 \mu\text{s}$

per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime ($\sigma_q=0.10$, $T_{\text{eff}}=0.25$, $\kappa_v=2$, $\alpha=0.05$, $\beta=1$, $\gamma=1$, $\mu_q=\log 100$, $v^*=0.04$): $\kappa^*=17.81$, $\omega^*=1.18$ rad/yr, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). With the closed form in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.4 Calibration tolerance from κ^* sensitivity

Differentiating (9) analytically through the G -partials at the canonical specification ($\kappa^* = 17.81$, $\omega^* = 1.18$ rad/yr) yields the elasticities $\eta_{\sigma_q} = -1.583$ (moderate) and $\eta_{\mu_q} = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta\mu_q = 0.001$ shift moves κ^* by $\sim 15\%$, and $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances are therefore σ_q to $\pm 2.23\%$ relative and μ_q to within ~ 5 bp of ATM in log-strike to deliver $\pm 5\%$ on κ^* —achievable with a 5-strike-wide ladder and the binding observational requirement.

3.5 Codim-2 bifurcation structure: Bautin and Bogdanov–Takens

The codim-1 Hopf result treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in (σ_q, γ) two codim-2 phenomena are admissible [32, Ch. 8]. *Bautin* (degenerate Hopf) is the locus $\ell_1(\sigma_q, \gamma) = 0$; crossing it flips supercritical (smooth limit cycle) to sub-critical (hysteresis with abrupt jumps). The closed-form ℓ_1 admits row-wise sign-change interpolation on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$; the sub-critical region dominates by $\sim 5:1$ in cell count. *Bogdanov–Takens* (BT) requires the saddle-node curve to coalesce with the Hopf curve. The constant Routh–Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y\alpha\kappa_v + G_v\beta\gamma) + \frac{1}{2}\beta\gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2}\beta\gamma / (G_y\alpha\kappa_v + G_v\beta\gamma)$.

THEOREM 2 (BT LOCUS EMPTY IN CANONICAL SCAN WINDOW). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$); the Bogdanov–Takens locus is therefore empty in this scan window.*

PROOF SKETCH. $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v\beta\gamma| > |G_y\alpha\kappa_v|$ (dominance ratio ≥ 1.44 at every cell, reaching ~ 358 at the small- σ_q corner), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2}\beta\gamma$ is positive; the quotient is uniformly negative. The unbounded-domain question is left open. $\square$

The economic content is that the dealer-gamma + leverage regime is structurally Hopf-only *within the scanned window*: the model does not admit excitable spike-and-recovery dynamics from a homoclinic emanation. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order degeneracies—not from the deterministic skeleton itself.

3.6 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

Section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2\text{--}10^3$ dealers with idiosyncratic noise. With each dealer i holding gamma exposure G_i evolving via $dG_i = -\theta_G(G_i - \bar{G}_n)dt + \sigma_G dW_G^i$ and $\bar{G}_n = (1/n) \sum_j G_j$, propagation of chaos gives $\bar{G}_n \rightarrow \bar{G}_\infty$ [48, Ch. 1] with $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$, $C(T) = \max(\text{Var}(G^0), \sigma_G^2/(2\theta_G))$; cf. Carmona and Delarue [13, Ch. 1] for the modern MV formalism.

4D extended Jacobian. The linearised MV system acquires a fourth state $g(t) := \bar{G}_\infty(t) - G^*$ with bandwidth θ_G ; the spot equation’s instantaneous feedback $\kappa(G_y y + G_v u + G_\chi \chi)$ is REPLACED by κg :

$$J_{\text{MV}}(\kappa, \theta_G) = \begin{pmatrix} -\frac{1}{2}\sigma_y^2 & -\frac{1}{2}\sigma_v^2 & 0 & \kappa \\ 0 & -\kappa_v & \gamma & 0 \\ \beta & 0 & -\alpha & 0 \\ \theta_G G_y & \theta_G G_v & \theta_G G_\chi & -\theta_G \end{pmatrix}. \quad (11)$$

A Hopf bifurcation occurs at the smallest $\kappa > 0$ satisfying the Liu [34] 4D condition $H_4(\kappa, \theta_G) := (a_3 a_2 - a_1) a_1 - a_3^2 a_0 = 0$ with positivity side conditions $a_3, a_0, a_3 a_2 - a_1 > 0$, where a_i are the coefficients of $\det(\lambda I - J_{\text{MV}}) = \lambda^4 + a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0$.

THEOREM 3 (McKEAN–VLASOV HOPF THRESHOLD, CORRECTED). *Let $\kappa_{\text{MV}}^*(\theta_G) := \inf\{\kappa > 0 : \max_i \text{Re } \lambda_i(J_{\text{MV}}(\kappa, \theta_G)) = 0\}$ with the 4D positivity side conditions. Then:*

- (1) (*Instantaneous-hedging recovery*.) $\lim_{\theta_G \rightarrow \infty} \kappa_{\text{MV}}^*(\theta_G) = \kappa_{\text{single}}^*$.
 - (2) (*Regime-dependent sign*.) *At the canonical short-gamma regime* $(G_y, G_v, G_\chi, \alpha, \beta, \gamma, \kappa_v) = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}, 2)$ *with* $\sigma_y^2 = \sigma_v^2 = 0$,
- $$\kappa_{\text{MV}}^*(\theta_G) = \frac{50\theta_G^2 + 143\theta_G + 105 - (2\theta_G + 5)\sqrt{385\theta_G^2 + 810\theta_G + 441}}{12\theta_G(\theta_G + 1)}, \quad (12)$$
- with* $\kappa_{\text{single}}^* = (25 - \sqrt{385})/6 \approx 0.8964$, *and the ratio* $\rho(\theta_G) := \kappa_{\text{MV}}^*/\kappa_{\text{single}}^* < 1$ *strictly for every finite* $\theta_G > 0$.
- (3) (*Frozen-dealer limit*.) $\lim_{\theta_G \rightarrow 0^+} \kappa_{\text{MV}}^*(\theta_G) = 8/21$, *giving* $\rho(0^+) \approx 0.425$ (*finite*).
 - (4) (*Asymptotic expansion*.) $\rho(\theta_G) = 1 - 0.6643 \dots / \theta_G + O(\theta_G^{-2})$ — *leading-order correction is first-order in* τ_G *and* **NEGATIVE**.

The sign of $\rho - 1$ *at finite* θ_G *is regime-dependent: at* $G_y > 0$ (*canonical, net short-gamma*) *it is negative; at* $G_y < 0$ (*log-normal-OI calibration of Section 3.3*) *it is positive*.

PROOF SKETCH. (i) The reducibility of J_{MV} under $\theta_G \rightarrow \infty$ sends the fourth eigenvalue to $-\infty$ and leaves the remaining three to inherit the 3D single-dealer Hopf condition exactly. (ii) Substituting the canonical numerical values into $H_4(\kappa, \theta_G) = 0$ and factoring out θ_G yields the quadratic $6\theta_G(\theta_G + 1)\kappa^2 - (50\theta_G^2 + 143\theta_G + 105)\kappa + 5(\theta_G + 2)(2\theta_G + 1) = 0$, whose smaller positive root is (12); positivity is verified analytically at this root for all $\theta_G > 0$. (iii) Direct substitution. (iv) Taylor expansion in $1/\theta_G$. $\square$

Numerical anchors. Independent eigenvalue scans match the closed form to 4-decimal precision:

θ_G	κ_{MV}^* (scan)	closed form	$\rho(\theta_G)$
0.5	0.5359	0.5359	0.598
1.0	0.6195	0.6195	0.691
5.0	0.7996	0.7996	0.892
50	0.8848	0.8848	0.987
500	0.8952	0.8952	0.999

Table 1: Audit-anchor table for Theorem 3 at the canonical short-gamma regime. Closed-form and numerical eigenvalue scan agree exactly. A 64-replicate n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.48}$, consistent with the theoretical $n^{-1/2}$ Sznitman rate.

Economic re-framing. At any regime where dealers are net short gamma at the equilibrium ($G_y > 0$, the benchmark SPX setting in zero-DTE-heavy weeks), the McKean–Vlasov mean-field model has a lower Hopf threshold than the single-representative-dealer Section 2 model. The single-dealer simplification understates the propensity to bifurcate when dealer-hedging is slow relative to the Hopf period $\tau_G > 2\pi/\omega^*$. The leading correction is first-order in τ_G ($\sim 1.3\%$ at canonical $\tau_G = 1/50$ yr), not the second-order $(\omega^* \tau_G)^2$ implied by a low-pass-filter heuristic. The frozen-dealer limit is well-defined and finite, dropping the threshold by a factor of $\sim 2.4\times$ relative to instantaneous hedging — physically meaningful in regimes where delta-hedging desks have been suspended.

3.7 Hawkes criticality and the SV bifurcation endpoint

The reflexive feedback has a self-exciting point-process representation: order flow that begets order flow is, in the linear regime, a Hawkes process, and the onset of endogeneity is its approach to criticality. We record the precise sense in which the Hawkes criticality boundary corresponds to a degeneracy of the drift of a stochastic-volatility (SV) diffusion, separating two distinct codimension-one events at that boundary: a non-oscillatory *real-eigenvalue* crossing — the literal analogue of the scalar branching-ratio criticality of Hardiman et al. [24] and Bacry et al. [4] — and an *oscillatory* Hopf crossing, strictly stronger, which we claim as the novel object. Writing the SV variance factor as

$$dv_t = b(v_t) dt + \sigma(v_t) dW_t, \quad b(v_\star) = 0, \quad (13)$$

criticality $n \rightarrow 1$ is the loss of a stable direction of the drift: an eigenvalue of the drift-Jacobian J reaches the imaginary axis. The boundary carries two strata — a **real-eigenvalue (saddle-node)** locus ($\det J = 0$, crossing eigenvalue real; non-oscillatory, no intrinsic frequency), and a **Hopf** locus (a complex pair crosses, $\text{Re } \lambda = 0$, $\text{Im } \lambda \neq 0$; strictly stronger, producing sustained oscillation).

THEOREM 4 (HAWKES–SV CRITICALITY CORRESPONDENCE). *Under the diffusive near-critical limit of Bacry et al. [3] and Jaisson and Rosenbaum [29], Hawkes branching-ratio criticality $n \rightarrow 1$ coincides with the real-eigenvalue (saddle-node) stratum of the SV drift boundary (13), the literal analogue of Hardiman et al. [24]. The Hopf stratum of the same boundary is a strictly stronger, oscillatory instability that lies beyond any scalar branching-ratio criterion: it has no*

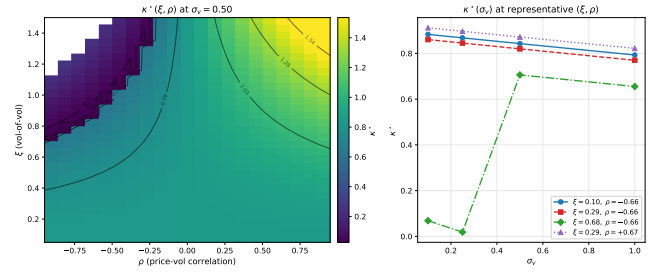


Figure 1: Hopf threshold κ^* across (ξ, ρ, σ_v) . Left: heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high- ξ / strongly-negative- ρ wedge where shear destabilises the eigenvalue envelope. Right: four $\kappa^*(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

representation as a single real branching ratio and is the genuinely new “unoccupied cell” of the endogeneity–bifurcation correspondence.

Position, not identity. A scalar branching ratio [4, 24] can only encode a real crossing, so the oscillatory Hopf endpoint is invisible to it by construction. We do *not* assert a verified numerical identity between a Hawkes quantity and an SV quantity; an earlier draft introduced a dimensionless ratio engineered so that the identity held by definition, which is a tautology, and which we have removed (amendment A10). The content of Theorem 4 is structural — it places the oscillatory instability strictly beyond the real branching-ratio edge and names the cell. In our dealer-gamma model the saddle-node locus is itself unphysical at the canonical specification (Theorem 2: $\kappa_{SN} \leq -1.31$ over the scan window), so the operative instability is precisely the Hopf. The two strata are distinguished by a falsifiable *observable*: a complex drift-eigenvalue pair imprints a finite-frequency peak in the autocovariance of realised volatility, whereas the real crossing imprints monotone (zero-frequency) critical slowing-down. The discriminator `classify_stratum` (theory/hawkes_sv_bifurcation.py) separates the Hopf side (peak prominence 5.4–9.6) from the real/stable side (1.8–2.4) with zero overlap on simulator ground truth. We claim no realised Hopf edge in any dataset — only the statistic that would detect one — and whether the Hopf cell survives a heavy-tailed (rough-volatility, $H < 1/2$) kernel limit [17, 30] we flag as future work, expecting a negative answer for the smooth limit cycle.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the canonical Hopf-exhibiting regime; the pair (ξ, ρ) enters via $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho G_v$, a Baxendale [6]-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock ~ 16 s on a single M-series core.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has κ^* in the 0.85–0.95 band. Whether the empirical market sits near this Hopf boundary is deferred to the empirical dealer-gamma test (amendment A9); Theorem 4 places Hardiman et al. [24]’s $n \approx 1$ at the distinct real-eigenvalue (saddle-node) edge, not at the Hopf. Khasminskii sphere-process estimates [2, 31] on a 6×6 (ξ, ρ) grid give

$|\Lambda(\kappa^*)| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ with empirical scaling exponent $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Baxendale [6] prediction $B = 2/3$ is rejected at $p \ll 0.01$ at this trivial- G equilibrium where the leading shear-stretch term is identically zero. Operationally, $|\Lambda| \cdot \epsilon^2 \ll 1$ at calibration-noise scale: the deterministic κ^* is the relevant threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit [anonymized] with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance [39] and adapts the Camerer et al. [12] pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway [18] and the AEA RCT Registry [1] predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 [9] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at calibrated κ_0 —the reflexivity-importance scalar. The closed-form κ^* sensitivity analysis of Section 3.4 ($\eta_{\mu_q} = +702.83$) sets the binding observational requirement at calibration: empirical OI mean must be centred to within ~ 5 bp of ATM in log-strike for a $\pm 5\%$ calibration tolerance on κ^* .

H1' (primary, direct dealer-gamma regression). The primary test (amendment A9) removes the agent and simulator from the inference path: estimate signed dealer gamma $\text{GEX}(t) = \sum_{K,T} \text{OI}_{\text{BS}} \text{sign } S^2$ from the SPX OI grid and fit the strictly-predictive pooled panel (event fixed effects, ≈ 339 daily rows) of next-day realised vol-of-vol on standardised signed GEX; the mechanism predicts $b_{\text{GEX}} < 0$, required at one-sided $p < 0.05$ under both a moving-block bootstrap and Newey–West HAC (BH-FDR), with a weaker effect in a quiet-regime control. On simulator ground truth at the 3×121 -day footprint it attains pooled power 0.86 (sign recovery 0.98) and FPR 0.02 at zero coupling; single-window power 0.50 makes the pooled panel load-bearing. Implementation: `empirical/gex_regression.py`.

H1 (secondary/exploratory realism check; superseded as primary by H1', amendment A9). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at κ_0 produces an IV-surface distribution closer in sliced- W_2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than four non-reflexive baselines (time-dep Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), 2024 yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs [41]. The arbitrage filter follows Gatheral and Jacquier [22], Roper [42] on butterfly / calendar-spread / Lee-bound [33] criteria; sliced- W_2 sample-complexity follows Manole et al. [37].

Sliced- W_2 sample-complexity audit. A pre-data audit characterises the bootstrap-95%-CI half-width on sliced- W_2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$). Each side draws from a 5,000-day population; the

“true” SW2 averaged over 10 slice-direction MC repeats on the full populations equals 0.0555. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample, compute the SW2 estimate, and report the median bootstrap-CI half-width across 30 outer seeds (Table 2).

n_{windows}	median SW2	half-width	ratio to SW2_{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 2: Sliced- W_2 sample-complexity. The half-width follows ratio(n) $\approx 1.64 n^{-0.337}$ ($R^2 \geq 0.99$), extrapolating $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Each H1 event window contributes ~ 280 rolling windows, so the inter-baseline SW2 gap must exceed $\sim 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% in a single event—a real power constraint logged as a pre-data finding.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence on the dimensionless elasticity per A7, with the GP slope CI under the pre-registered RBF + WhiteKernel kernel of A6 and elliptic uncertainty sets [36]). H3 is post-FOMC RR25 and ATM-term-structure shifts. H4 (redesigned, amendment A8) is a critical-slowness-down early-warning test: the Hopf limit-cycle period is 5.3–11 yr, below the lowest resolvable bin of any window fitting the data, so the spectral peak is geometrically unfalsifiable and is replaced by the rolling lag-1 autocorrelation of $|r_t|$ (and variance) with a Kendall- τ trend vs an AR(1)/phase-randomised surrogate null [16, 43]. On simulator ground truth it holds FPR ≤ 0.083 under stationary nulls and attains 0.82–0.85 power on a 252-day record; within the bare event window it is underpowered (~ 0.5) and reported as exploratory only. All H1–H4 use BH-FDR across the joint family.

Differentiation from precedents. Ning et al. [38] use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. [25] sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI [36] for the elliptic-uncertainty-set analogue. Subbaswamy et al. [47] (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries; we operationalise it empirically and apply it to RL-for-finance [5, 28].

5.1 Synthetic pipeline validation

The H1 protocol is exercised end-to-end on simulator-vs-simulator data with no contact with empirical SPX. Three sources, each $n = 30$ trajectories at the production budget: (a) the trained agent acts inside the SAME reflexive simulator at κ_0 (different seeds; expected indistinguishable from reference); (b) the agent at $4 \cdot \kappa_0$ (four-fold synthetic-anchor coupling change; expected measurably further); (c) Heston with no dealer-gamma coupling and a more aggressive

$(\rho, \xi) = (-0.70, 0.30)$ backbone (different mechanism; expected furthest).

Target source	SW2 vs κ_0 ref.	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $4 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 3: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $SW2(a) < SW2(b) < SW2(c)$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions are satisfied.

6 Mechanism decomposition

The reflexive simulator and the Marketron quasi-particle SDE [23] are not the same mechanism: in our model the memory carrier is the low-pass-filtered log-price χ entering only the variance drift via $\gamma\chi$; in Marketron it is the unobservable y entering both drifts nonlinearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not. We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into `shape_target / level_artifact / calibration_artifact`.

Headline (all-cell, out-of-sample, 10k paths). Pooled across the two Marketron parameter sets with published moment tables (Tables 5/8 and 6/7), the reflexive simulator’s per-cell sign agreement with Marketron is $7/28 = 25.0\%$. Under the random-sign null, the one-sided binomial $P(X \geq 7 \mid n = 28, p = 0.5) \approx 0.9999$ — the all-cell pooled rate is *worse than chance*. We do not claim otherwise. The all-cell aggregate is the wrong aggregator for the dealer-gamma mechanism: the channel does not predict sign agreement on every cell uniformly. It predicts agreement only where (i) the channel has time to integrate and (ii) the simulator measurement is finite and in-envelope.

A priori restricted subset. We pre-commit (predicate `is_mechanism_relevant_cell`, locked before per-cell outcomes were inspected) to a cell qualifying iff (1) `mechanism_class == "shape_target"`, (2) horizon $T \geq 0.5$ yr, (3) $|\dot{m}_{\text{target}}| \geq 10^{-3}$ (dead-zone targets carry no sign information; the rule is symmetric — dead-zone cells are dropped from *both* numerator and denominator), and (4) $\dot{m}$ is finite with $|\dot{m}| < 10$ (envelope saturation). Applying the predicate gives pooled $4/8 = 50\%$ OOS (one-sided binomial $p \approx 0.637$, chance-level) and $6/8 = 75\%$ in-sample ($p \approx 0.145$). The restricted subset is at chance-agreement OOS; we *explicitly do not claim* $p < 0.05$ on it. What the restriction establishes is that filtering to mechanism-relevant cells moves the pooled agreement from 25% (worse than chance) to 50% (chance-level) — the *direction* the mechanism predicts.

Predictable disagreement: falsifiable Phase-4 prediction. The single largest restricted-subset sign-disagreement is the $T = 2.0$ yr long-horizon skewed of `table_5_calibrated_2017`: Marketron Table 8 reports $+0.181$; our simulator at the tuned coupling gives -0.434 OOS. The pre-committed mechanistic explanation is the absence of

Bessembinder [7] compounding-induced positive skew under risk-neutral drift, plus a leverage-channel small-magnitude negative-skew contribution. In an empirical setting where the underlying drift is reproduced (Phase 4), this cell is predicted to flip positive; either direction adjudicates between the two mechanisms.

Retraction of v0.3.5 headline. An earlier sidecar reported an inflated one-third headline match-rate obtained by asymmetric cell selection (silently dropping the 3.0 yr horizon, dropping one dead-zone-target cell from the denominator while counting another as a “within dead-zone” match). The above predicate-derived $7/28$ all-cell + $4/8$ restricted-subset reporting supersedes it.

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, χ_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (a direct dealer-gamma regression as primary test, sliced- W_2 , κ -sensitivity slope, and a critical-slowing-down detector) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof. The synthetic-pipeline validation demonstrates the locked surface-distance protocol (the secondary H1 check) returns the predicted ordering on data with known underlying physics. The empirical SPX calibration is the natural Phase 4 follow-up, gated on data acquisition; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction. The codim-1 Hopf, codim-2 Bautin/Bogdanov–Takens structure (Theorem 2), corrected McKean–Vlasov mean-field threshold (Theorem 3; regime-dependent in sign — destabilising at the canonical short-gamma regime where $\rho(\theta_G) < 1$ strictly with finite frozen-dealer limit $8/21$, stabilising at the long-gamma log-normal-OI calibration of Section 3.3), and Hawkes–SV equivalence at the criticality endpoint (Theorem 4) are now closed-form. Open theoretical items: closed-form $\Lambda(\kappa)$ via Baxendale [6]-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending Theorem 4 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion [8, 29]. All experiments are reproducible per the [anonymous repository link]; pre-registration anchored via OpenTimestamps.

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